

Ahmer Alam

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Education

King's College London | Computer Science BSc with a Year in Industry | Sep 2022 - Jul 2026

- Grade: First Class Honours (82%)

Dr Challoner's Grammar School | A-Level | Sep 2020 - Jun 2022

- Mathematics: A* • Physics: A* • Computer Science: A* • Extended Project Qualification: A
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Work Experience

Software Engineer | Skyhigh Security | Aug 2024 - Sep 2025

- Contributed to the Neo-UI monorepository, delivering 110+ pull requests and 12K+ lines of Angular/TypeScript code across multiple web apps.
 - Independently led rework of End User Remediation app from AngularJS to Angular, rebuilding from scratch via a Test-Driven Development approach to reduce technical debt and loading times.
 - Implemented OAuth 2.0 Token Exchange in Java Spring Boot for their Classifications Middleware Service, enabling fine-grained access control through audience- and scope-specific tokens.
 - Designed functional and performance tests for their ML Text Categoriser model across 100+ multilingual datasets, integrating with Jenkins pipelines to automate execution, log artifacts, and visualise result plots.
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Projects

Reinforcement Learning for Risk Board Game | GITHUB, PAPER

- Designed a map-agnostic Risk engine for generalised gameplay and experimentation.
- Engineered a Proximal Policy Optimisation (PPO) pipeline within the OpenAI Gymnasium framework to train reinforcement learning agents, achieving >80% win rate against heuristic baselines.
- Adapted official SMG Studio "risk-dice" repository to precompute a BlitzProbabilityMatrix, replacing runtime stochastic simulation with lookup-based battle resolution to lower per-episode computation.
- Conducted controlled experiments across multiple map variants to study the effect of graph structure on strategy and game balance dynamics; documented results in an academic-style research paper.

Tennis ML Predictor | GITHUB

- Developed an end-to-end machine learning pipeline for ATP tennis match prediction, automating live data collection, model training, and evaluation.
- Processed and engineered features from 60K+ ATP matches spanning 20 years using Pandas and NumPy, deriving predictive metrics including Elo ratings, fatigue, recent form and head-to-head records.
- Trained and evaluated multiple ML models (Elo, Multi-Layer Perceptron, and XGBoost), achieving ~70% prediction accuracy on unseen Wimbledon 2025 matches, comparable to leading bookmaker odds.
- Exposed trained ML models through a FastAPI service for real-time ATP match predictions.

The World Blueprint: A Semi-Automated Content Creator | GITHUB, TIKTOK, YOUTUBE

- Founded an educational channel on history, culture, and science, achieving 80K+ views and 2K+ likes.
 - Built a scalable, semi-automatic pipeline combining AI-generated scripts, web-scraped images, and Canva/Excel bulk create tools to manufacture 200+ "Fact vs Fiction" shorts across seven topics.
 - Engineered Python automation leveraging YouTube and TikTok APIs to schedule and upload videos, enabling streamlined publishing with playlists and metadata.
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Technical Skills

Languages:

- Python • Java • JavaScript/TypeScript • Scala • SQL

Frameworks and Libraries:

- Angular • React • Spring Boot • NumPy • Pandas • Scikit-learn • PyTorch • FastAPI

Tools and Practices:

- Git • Jenkins • Docker • OAuth • CI/CD • Agile